Exploratory Data Analysis

OASIS-2 visit pairs: clinical metadata + structural MRI | Predicting next-visit cognitive decline

DSCI 4350, Group 5: Christina Dames, Cinthia Ochoa Torre, Rayna Alexandre-Petrie | Checkpoint 2

150
patients

223
visit pairs

1,367
MRI scans (373 sessions)

20
model features

13.9%
declined (target = 1)

Split	Patients	Visit pairs	Declined	Decline rate
Train	119	178	25	14.0%
Test	31	45	6	13.3%

Dataset file: oasis2_visit_pairs.csv | SHA-256: 8668b506a03b3181... | random_state = 42

Contents and Reading Guide

Sections follow the data from its sources to the modeling decisions it supports

1 Data Sources and Lineage

Data Lineage 3

Dataset at a Glance 4

2 Target

Target: Stable vs. Declined 5

3 Features

Feature Statistics 6

Feature Distributions by Class 7

Features vs. Target 8

Feature Correlations 9

Which Features Separate the Classes? 10

4 MRI Scans

MRI Inventory and Acquisition 11

Example Scans by Dementia Rating 12

MRI Intensity and Quality 13

MRI-Derived Measures vs. Target 14

5 Summary

Key Findings 15

Implications for Modeling 16

Appendix Dataset Card 17

How to read this report

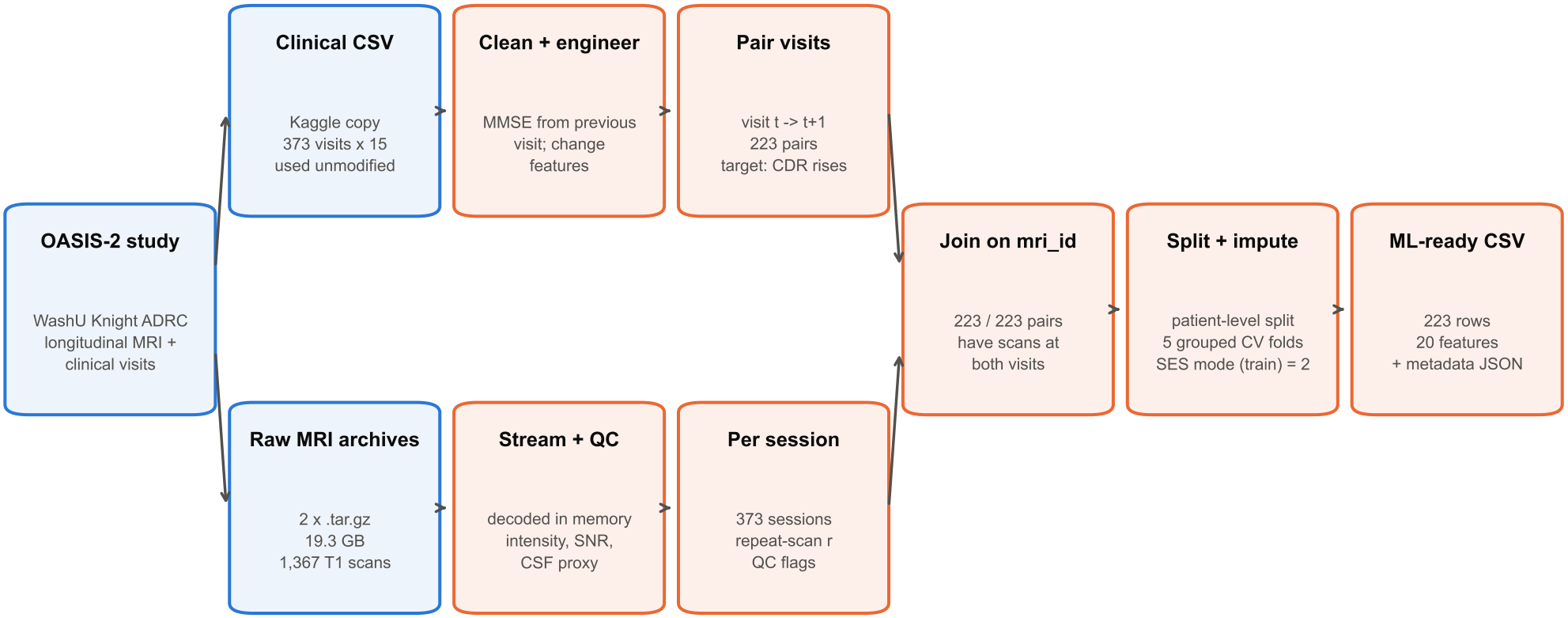
- Blue = stable (0), orange = declined (1) on every chart.
- Train AUC scores use training patients only; the test set stays unseen.
- Section 3 covers all model inputs; Section 4 covers the MRI scans.
- Figures are saved as reports/figures/<page>_<name>.png.

Glossary

Visit pair	One row: a patient's visit t (features) and next visit t+1 (target).
CDR	Clinical Dementia Rating: 0 none, 0.5 very mild, 1 mild, 2 moderate.
Declined	Target = 1 when CDR at visit t+1 is higher than at visit t.
MMSE	Mini-Mental State Examination, 0 to 30; lower = worse cognition.
nWBV	Normalized whole-brain volume (share of intracranial volume), from the CSV.
eTIV / ASF	Estimated total intracranial volume / atlas scaling factor (~1/eTIV).
SES	Socioeconomic status, 1 (highest) to 5 (lowest).
CSF proxy	Share of dark, fluid-like voxels in the center of the raw scan.
SNR	Signal-to-noise ratio of a scan (image quality).
Train AUC	ROC-AUC of one feature alone on training patients; 0.5 = no signal.

Data Lineage

Two OASIS-2 sources, one ML-ready file: clinical visits and raw MRI scans are joined on mri_id (src/data/)



Blue = external sources Orange = this project's transformations

- Top branch (src/data/make_dataset.py): MMSE filled from the patient's previous visit, backward-looking change features, each visit paired with the next.
- Bottom branch (src/data/make_mri_dataset.py): the 19 GB archives are streamed, not extracted (about 45 GB on disk); each scan gets quality and CSF measures.
- Excluded raw columns: Group (assigned from the full visit history, so it leaks the target) and Hand (constant).
- Imputation uses training patients only; every step is deterministic (random_state = 42) and SHA-256 hashes are stored in the metadata JSON files.

Dataset at a Glance

One row = one visit pair; the features describe the patient and the scan at visit t

Columns by group

Group	Source	n	Columns
Demographics	Clinical CSV	5	age, sex_male, educ, ses, ses_imputed
Clinical / cognitive	Clinical CSV	2	mmse, cdr
MRI measures (CSV)	Clinical CSV	3	etiv, nwbv, asf
Visit timing	Derived from CSV	4	days_since_baseline, days_to_next, has_prev_visit, days_since_prev
Change since previous visit	Derived from CSV	5	mmse_change_prev, nwbv_change_prev, cdr_change_prev, mmse_rate_prev, nwbv_rate_prev
MRI scan measure	Raw MRI scans	1	mri_csf_proxy
MRI quality (audit)	Raw MRI scans	4	mri_n_scans, mri_snr, mri_repeat_r, mri_qc_flag (not model inputs)
Target	CDR at t and t+1	1	declined (cdr_next is kept for analysis only, never as an input)
IDs and splits	-	5	subject_id, mri_id, visit, split, cv_fold

Source and output files

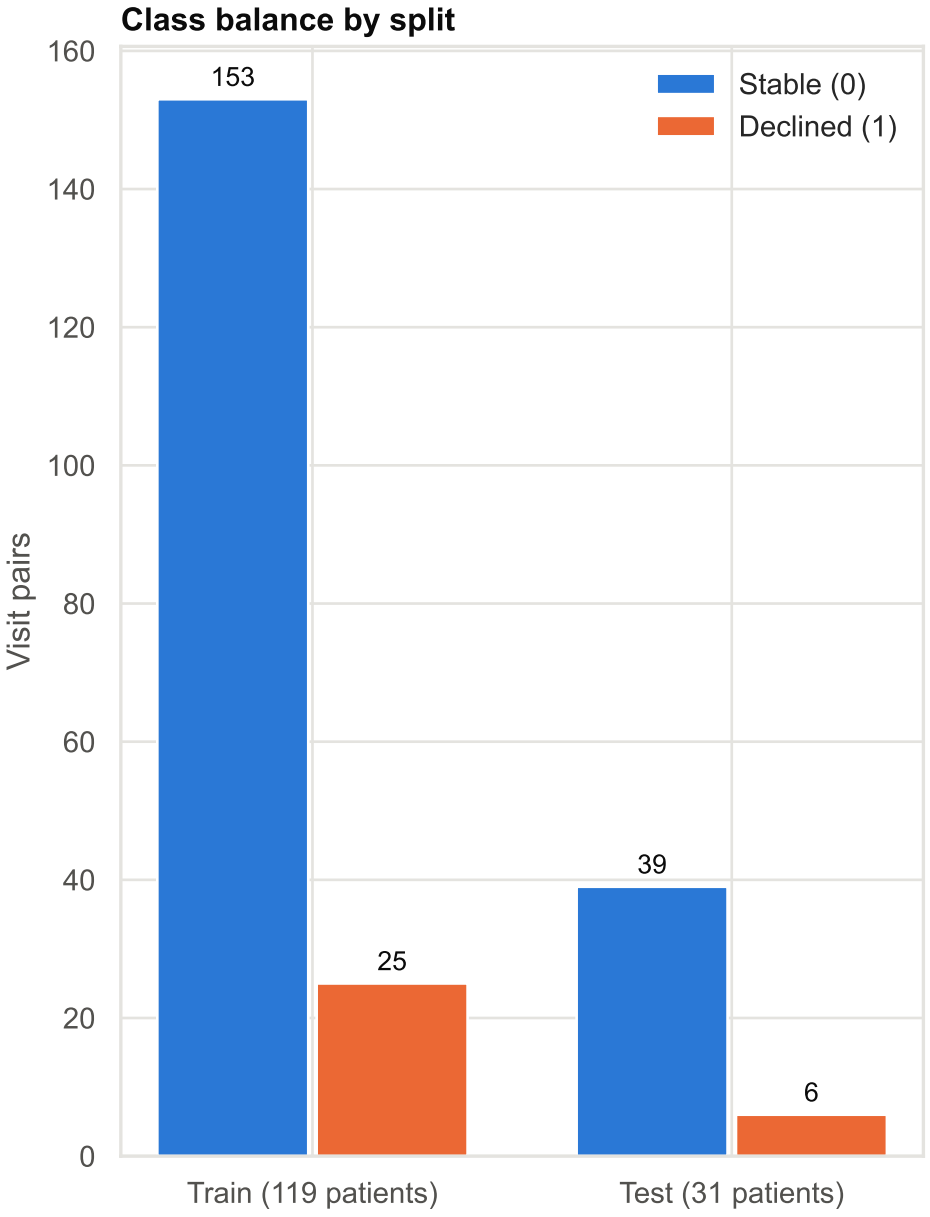
File	Origin	Records	Size	SHA-256
oasis_longitudinal.csv	Kaggle (OASIS-2)	373 visits	28 KB	d2f0a15ff35f
OAS2_RAW_PART1.tar.gz	oasis-brains.org	771 scans	10.9 GB	6df8fa570698
OAS2_RAW_PART2.tar.gz	oasis-brains.org	596 scans	8.4 GB	50bc39b6d911
oasis2_visit_pairs.csv	This project	223 pairs	42 KB	8668b506a03b

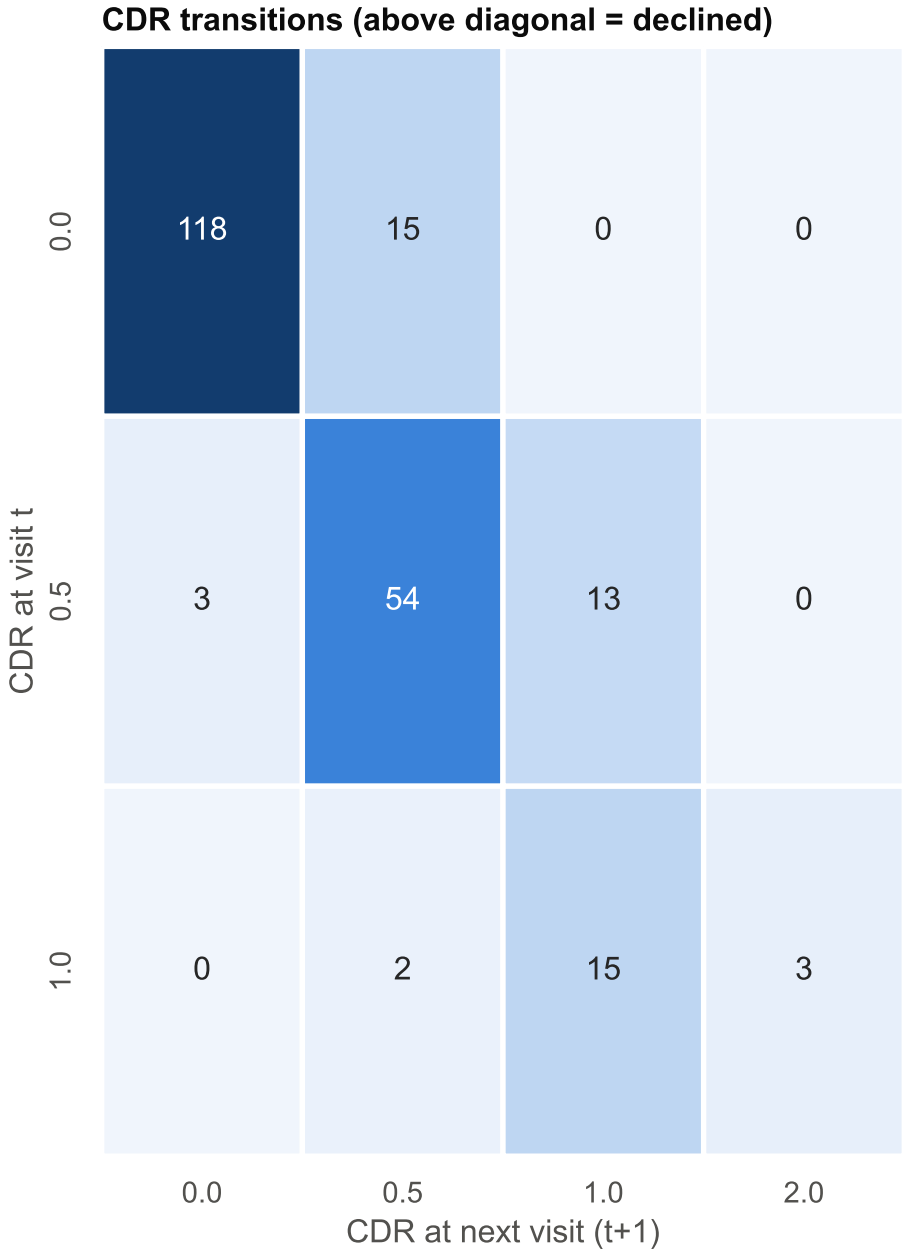
Linkage and split checks

Check	Result
Visits with an MRI session	373 / 373
Pairs with scans at t and t+1	223 / 223
Scans / sessions	1,367 / 373
Sessions flagged (QC)	19
Train / test patients	119 / 31
Patients in both splits	0

Target: Stable vs. Declined

Decline is rare (about 14%) and the patient-level split keeps that rate similar in train and test





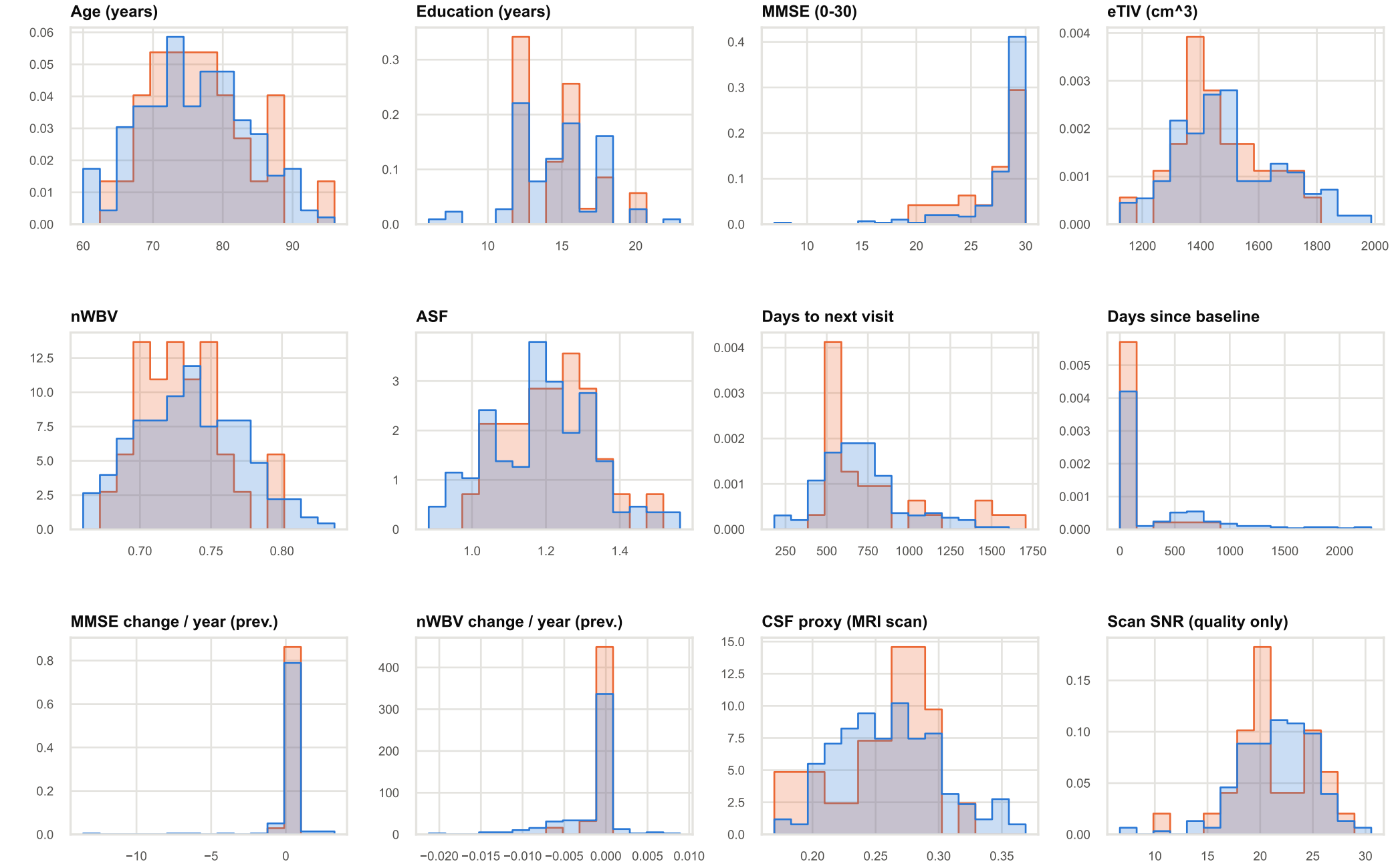
Feature Statistics

Mean (SD) by target class on all 223 pairs; univariate ROC-AUC on the training split only

Group	Feature	Range	Stable mean (SD)	Declined mean (SD)	Train AUC
Demographics	age	60 to 96	76 (7.55)	76.6 (7.27)	0.52
	sex_male	0 to 1	0.438 (0.497)	0.452 (0.506)	0.52
	educ	6 to 23	14.6 (2.94)	14.7 (2.57)	0.50
	ses	1 to 5	2.45 (1.12)	2.19 (1.05)	0.45
	ses_imputed	0 to 1	0.0365 (0.188)	0.129 (0.341)	0.53
Clinical / cognitive	mmse	7 to 30	27.9 (3.26)	26.9 (3.19)	0.39
	cdr	0 to 1	0.237 (0.327)	0.306 (0.334)	0.56
MRI measures (CSV)	etiv	1,123 to 1,987	1,494 (179)	1,468 (145)	0.51
	nwbv	0.66 to 0.837	0.735 (0.0366)	0.729 (0.0289)	0.46
	asf	0.883 to 1.56	1.19 (0.14)	1.21 (0.118)	0.49
Visit timing	days_since_baseline	0 to 2,288	308 (500)	79.5 (217)	0.39
	days_to_next	182 to 1,707	709 (248)	780 (339)	0.51
	has_prev_visit	0 to 1	0.359 (0.481)	0.129 (0.341)	0.40
	days_since_prev	0 to 1,554	244 (362)	79.5 (217)	0.40
Change since previous visit	mmse_change_prev	-12 to 5	-0.0521 (1.38)	0 (0.258)	0.49
	nwbv_change_prev	-0.034 to 0.011	-0.00263 (0.00623)	-0.000581 (0.00258)	0.60
	cdr_change_prev	-0.5 to 0.5	0.0156 (0.113)	0 (0)	0.49
	mmse_rate_prev	-13.6 to 3.26	-0.103 (1.33)	-0.00851 (0.162)	0.49
	nwbv_rate_prev	-0.0213 to 0.00891	-0.00147 (0.0036)	-0.000282 (0.00124)	0.60
MRI scan measure	mri_csf_proxy	0.17 to 0.369	0.261 (0.0406)	0.255 (0.0392)	0.48

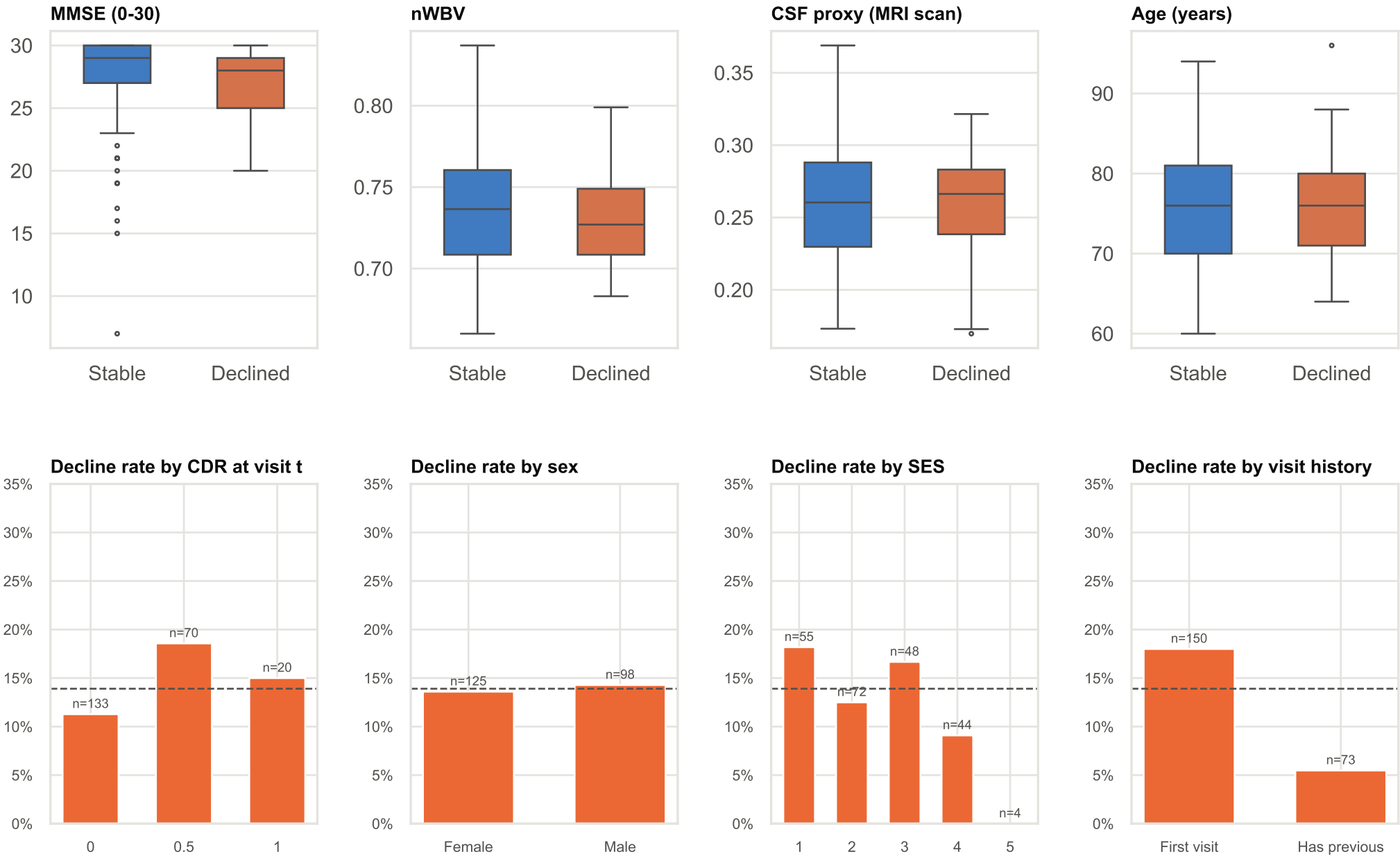
Feature Distributions by Class

Densities are normalized within each class so the small declined group stays visible



Features vs. Target

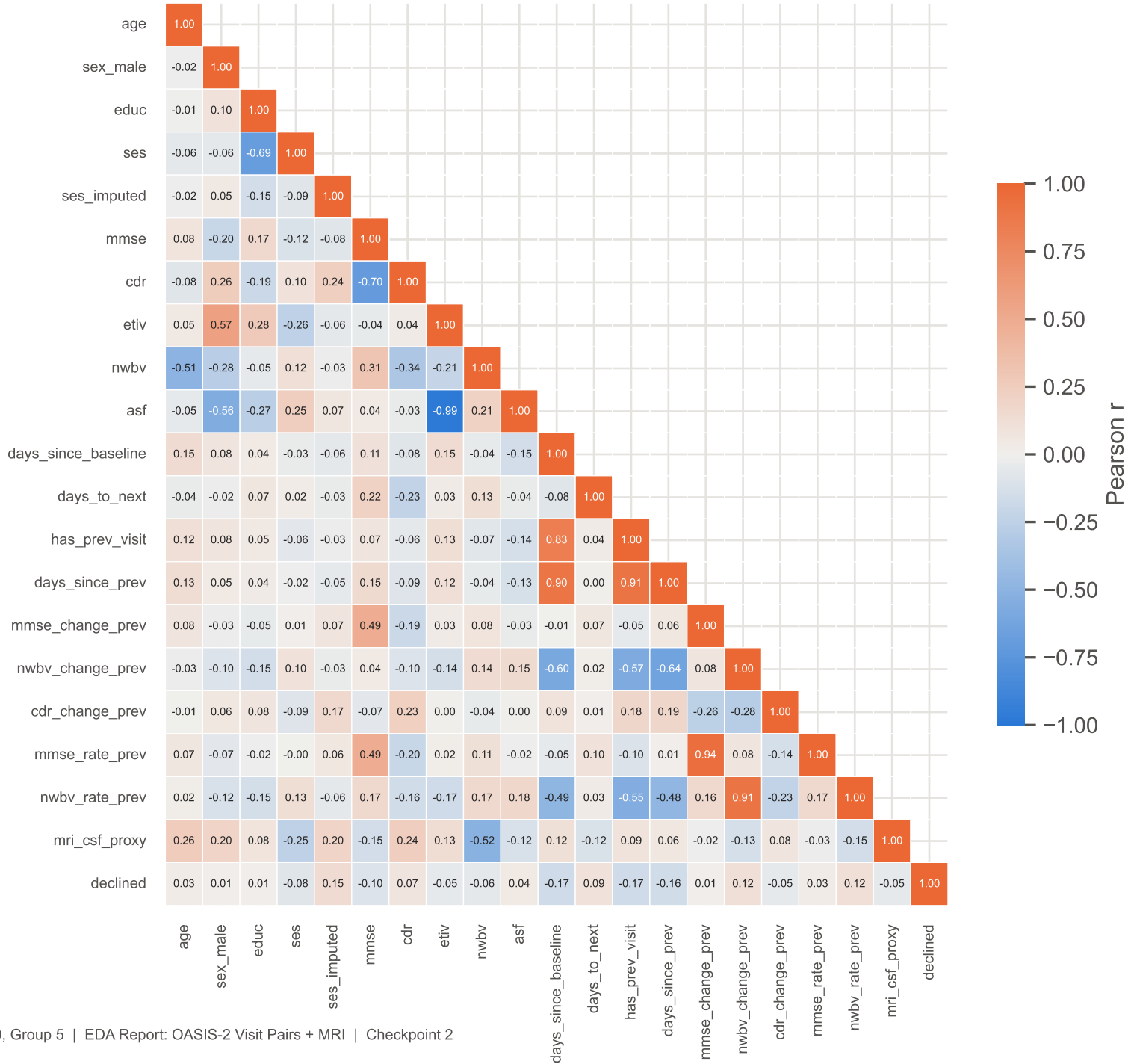
Cognitive scores and baseline CDR separate the classes more than brain volume, the MRI scan measure, or demographics



Dashed line = overall decline rate (13.9%). n = visit pairs in each group.

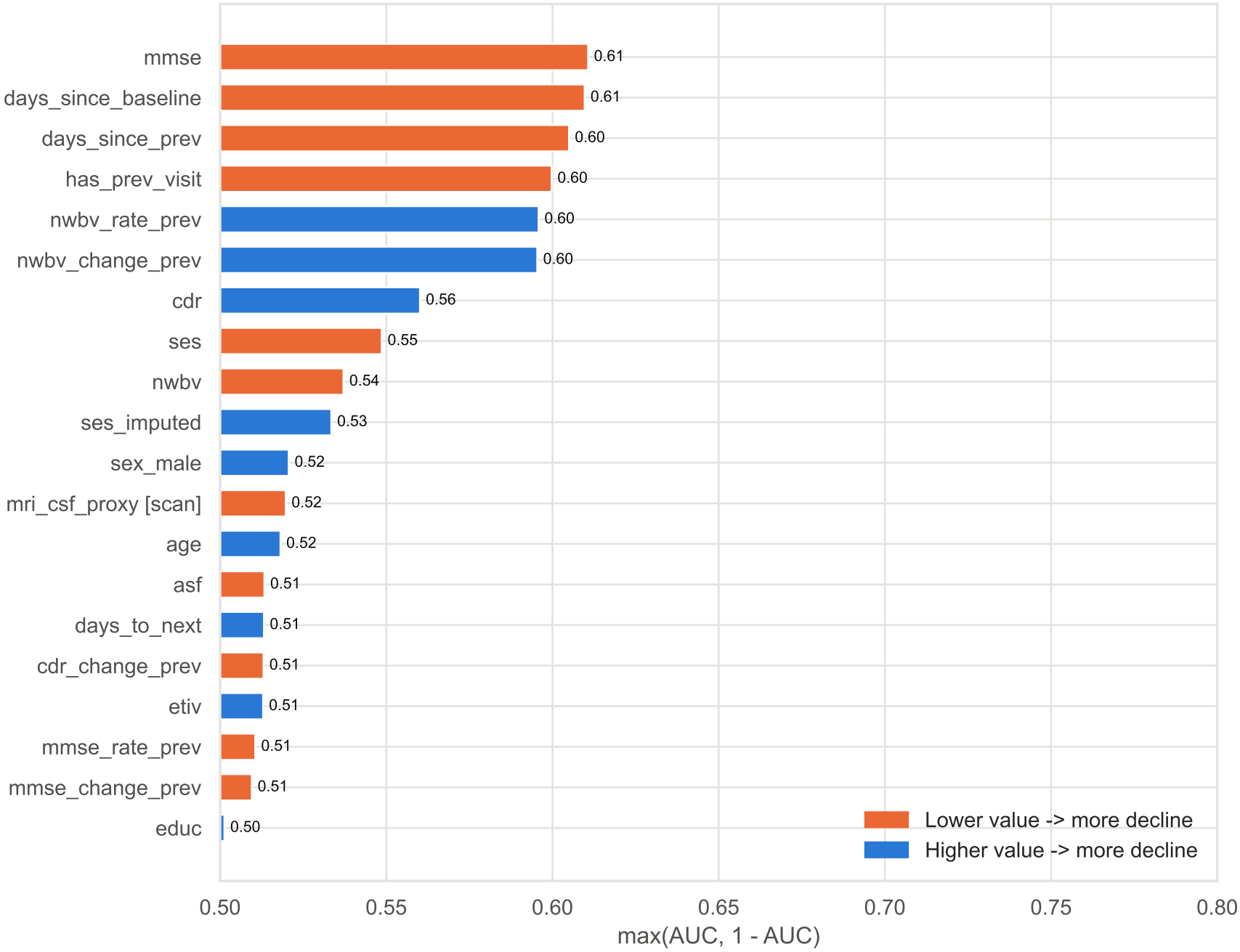
Feature Correlations

Pearson r, including the target. eTIV and ASF are almost perfectly redundant (r = -0.99); the CSF proxy tracks nWBV (r = -0.52)



Which Features Separate the Classes?

Univariate ROC-AUC per feature on the training split (0.5 = no signal). [scan] = computed from the raw MRI

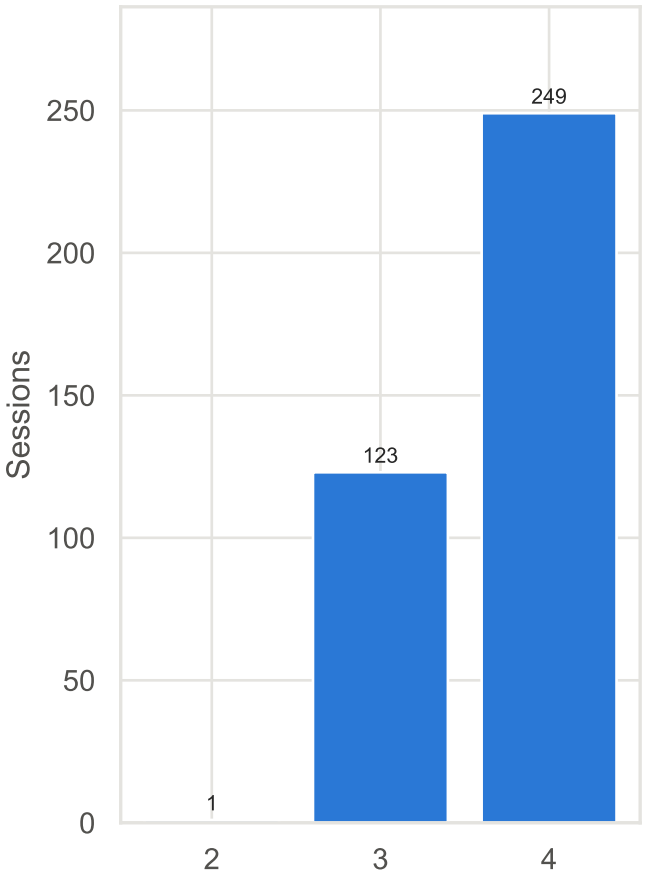


MRI Inventory and Acquisition

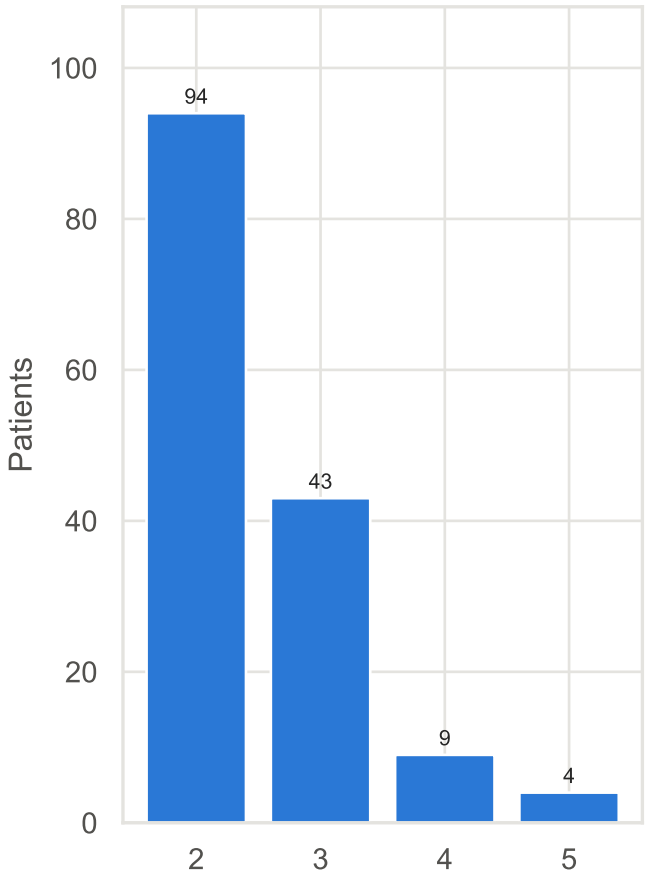
Every scan shares one acquisition format, so the volumes can be stacked without resampling

Dimensions (x, y, z)	Voxel size	Data type	Scans
256 x 256 x 128	1 x 1 x 1.25 mm	float32	1,367
Format: Analyze 7.5 (.img + .hdr)	T1-weighted MPRAGE, sagittal	big-endian	

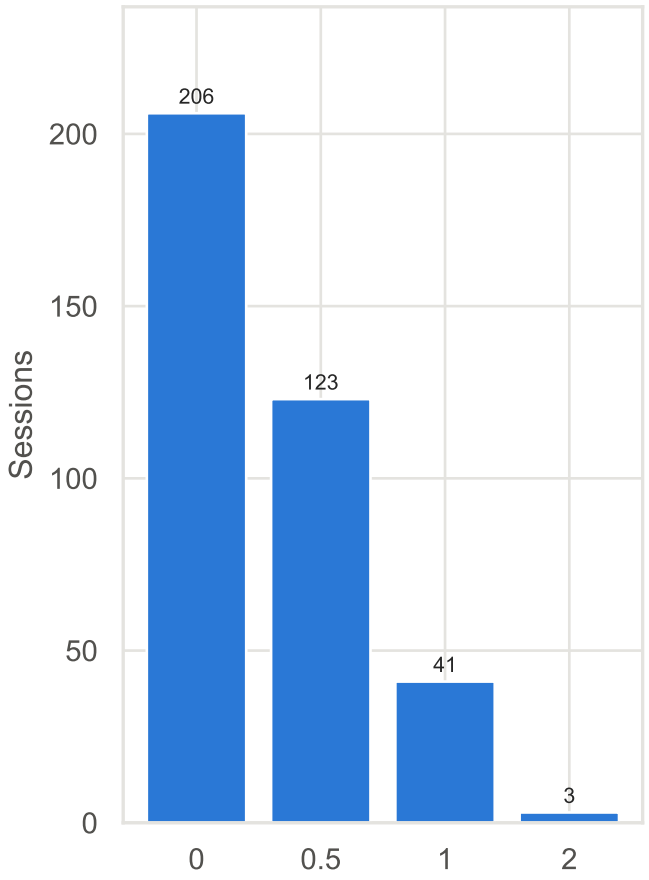
Repeated scans per session



MRI sessions per patient

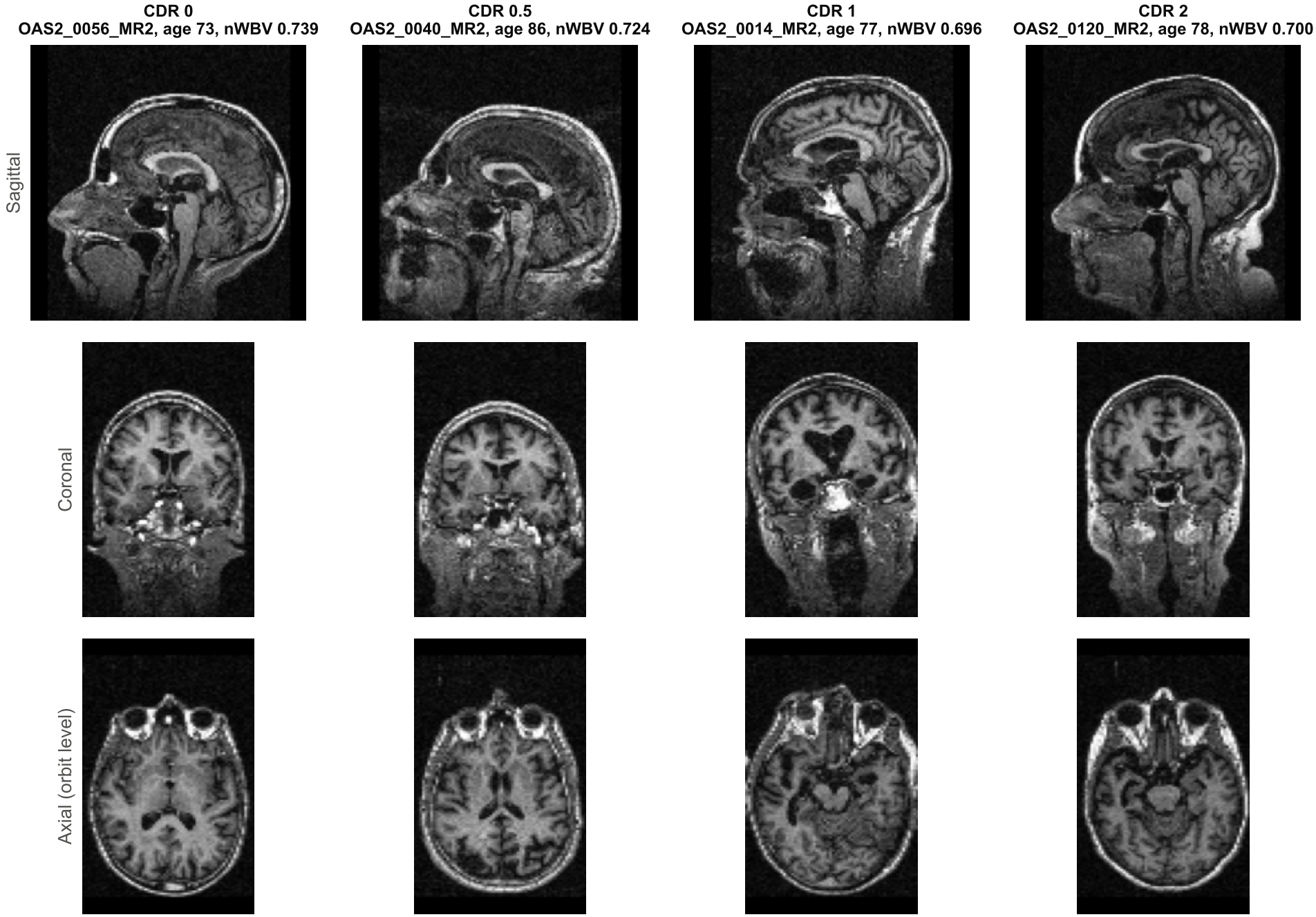


Sessions by CDR at scan



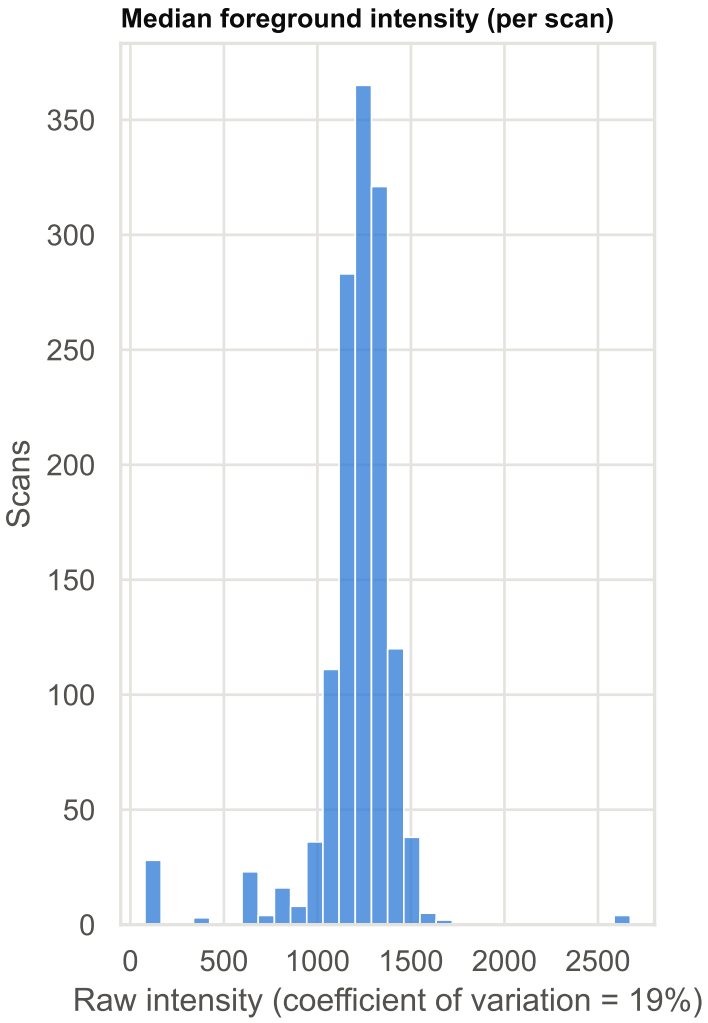
Example Scans by Dementia Rating

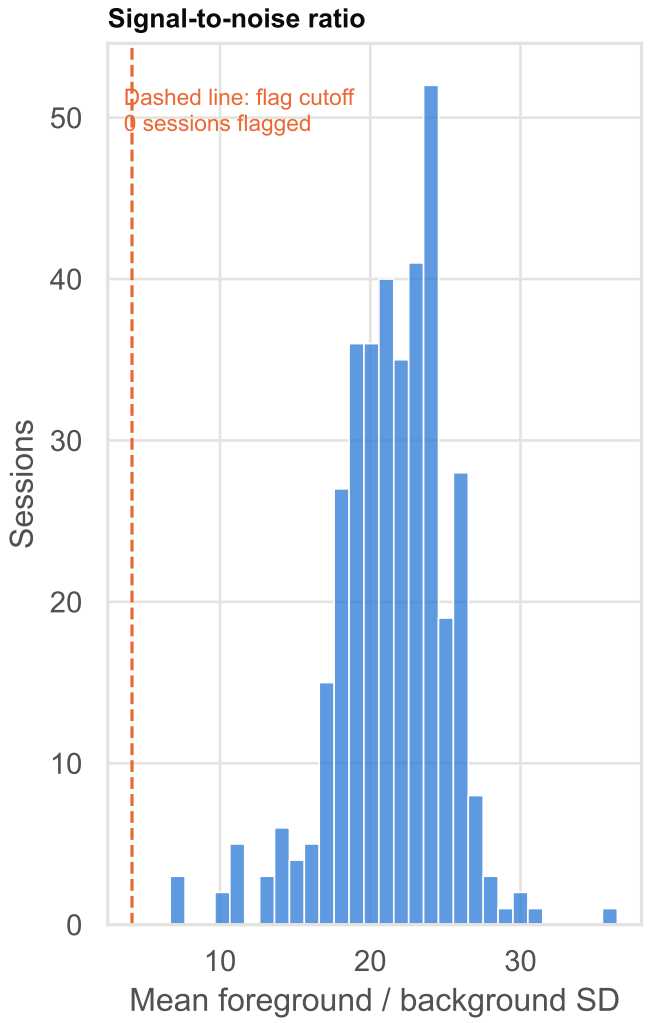
Central slices of the first scan in one session per CDR level (raw: not skull-stripped or registered)

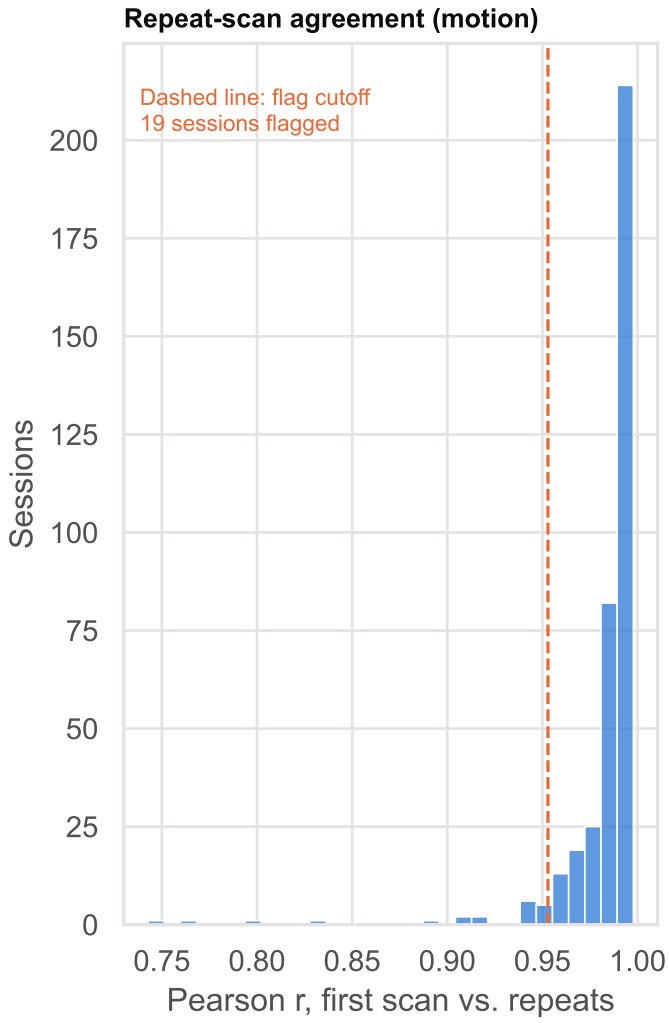


MRI Intensity and Quality

Raw intensity scales differ between scans; 19 of 373 sessions are flagged as quality outliers



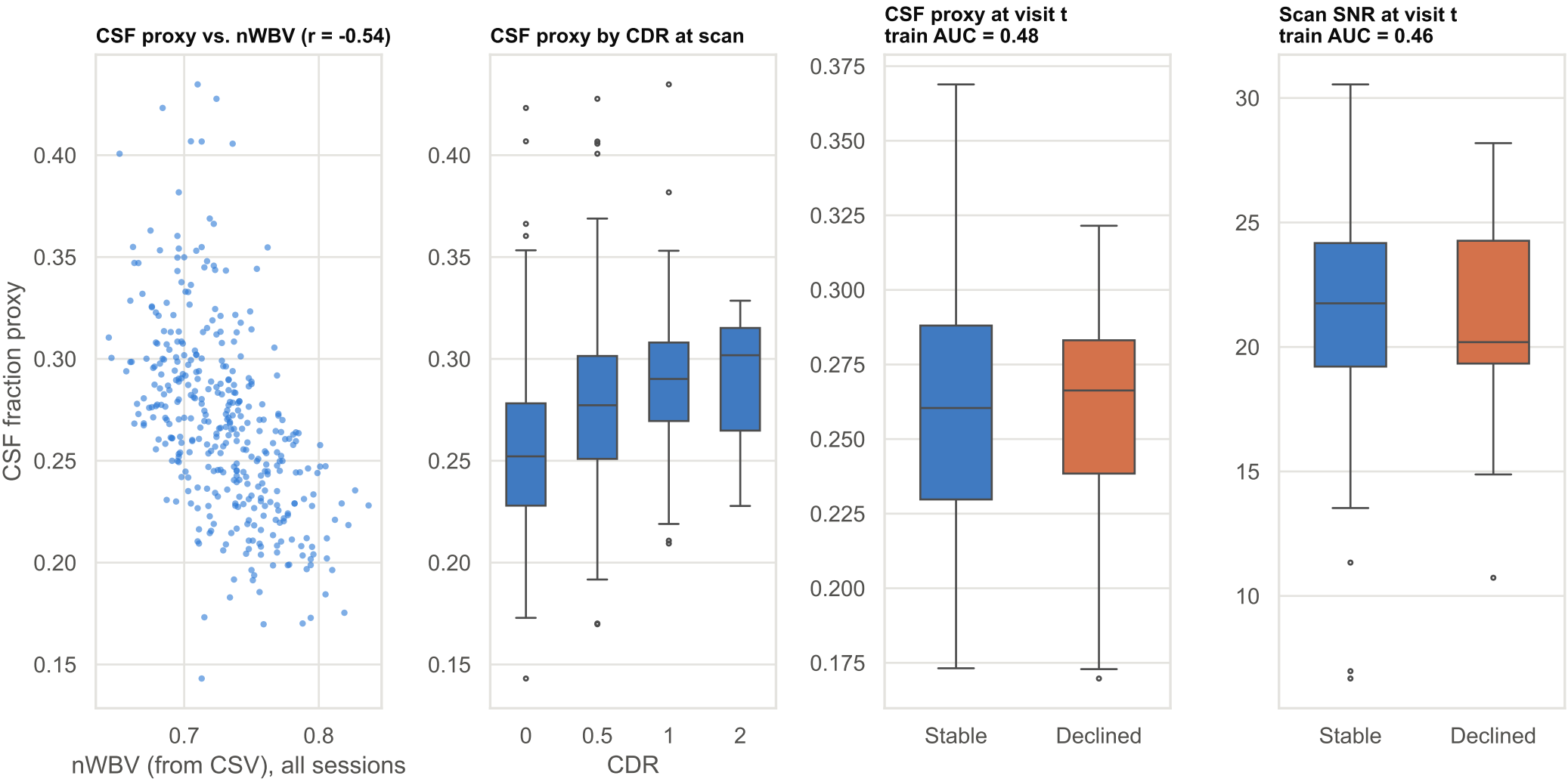




- Intensities are in arbitrary scanner units, so each scan must be normalized (e.g., z-score within the brain) before it is fed to a CNN.
- Flag = more than 3 IQR below the first quartile. Flagged sessions stay in the dataset (mri_qc_flag) for a sensitivity analysis.

MRI-Derived Measures vs. Target

The CSF proxy tracks atrophy and CDR, but, like nWBV, it separates future decliners only weakly



- For comparison, nWBV from the CSV has a train AUC of 0.46. AUC below 0.5 means lower values go with decline.
- SNR should not separate the classes; an AUC near 0.5 confirms that scan quality is not a hidden proxy for the outcome.
- The CNN is expected to extract richer structure (e.g., ventricle and hippocampus shape) than this single global measure.

Key Findings

What the clinical data and the MRI scans show, with every number computed from the ML-ready dataset

Data

- 150 patients, 223 visit pairs, and 1,367 MRI scans; 223 of 223 pairs have scans at both visits.
- Imbalanced target: 31 of 223 pairs declined (13.9%); the test set has only 6 positives.
- All scans share one format (256 x 256 x 128, 1 x 1 x 1.25 mm). Raw intensities vary between scans, and 19 sessions are flagged for motion.

Signal

- Strongest single features (train AUC): mmse (0.61), days_since_baseline (0.61), days_since_prev (0.60).
- Cognition is the clearest clinical signal: mean MMSE 26.9 for decliners vs. 27.9 for stable pairs; decline rate 19% at CDR 0.5 vs. 11% at CDR 0.
- Brain-volume measures are weak alone: nWBV train AUC 0.54; the CSF proxy from the scans tracks nWBV (r = -0.52) but reaches only 0.52.
- Most decline happens early in follow-up: 18% of first-visit pairs vs. 5% of later pairs.
- eTIV and ASF are redundant (r = -0.99). Scan SNR does not separate the classes (AUC 0.46), so image quality does not leak the outcome.

Implications for Modeling

How the findings shape the metadata-only, MRI-only, and multimodal models in Sprint 3

Evaluation

- Report recall, F1-score, and ROC-AUC (accuracy alone rewards predicting 'stable'); use class weights or resampling.
- Select models with the 5 patient-grouped CV folds; touch the test set once and report its metrics with their uncertainty.

The three models

- Metadata-only: the 19 CSV features; keep either eTIV or ASF for linear models.
- MRI-only: CNN on preprocessed scans: average the 2-4 repeats, skull-strip, register to a template, z-score intensities, and resample (e.g., 128^3 or 2D slices).
- Multimodal: combine the CNN embedding with the metadata features, using the same patient-level splits so results are comparable.

Risks to manage

- Small sample (223 pairs): prefer transfer learning or a small CNN with augmentation and early stopping.
- Re-run the final models without the QC-flagged sessions as a sensitivity check.

Dataset Card: OASIS-2 Visit Pairs with MRI (ML-ready)

Summary

An ML-ready longitudinal dataset for predicting next-visit cognitive decline in older adults. Each row pairs one clinical/MRI visit with the same patient's next visit. Features describe the patient and their MRI scan at the earlier visit; the binary target declined is 1 when the Clinical Dementia Rating (CDR) increases at the next visit. Every row links to the raw structural MRI scans of both visits through mri_id.

Property	Value
Version	2.0 (Checkpoint 2: clinical metadata + MRI scans)
Instances	223 visit pairs from 150 patients
Features	20 model inputs: 19 clinical metadata features + 1 MRI scan measure (see Fields)
MRI quality columns	4 audit columns (scan count, SNR, repeat-scan agreement, QC flag); not model inputs
Target	declined (0 = stable, 1 = declined); 31 positives (13.9%)
Splits	Patient-level train (178 pairs, 119 patients) / test (45 pairs, 31 patients) + 5 grouped CV folds in train
Linked MRI	1,367 raw T1 scans in 373 sessions; all 223 pairs have scans at both visits
Files	data/processed/oasis2_visit_pairs.csv, oasis2_mri_sessions.csv, oasis2_mri_scans.csv (+ metadata JSON for each build)
Curators	Christina Dames, Cinthia Ochoa Torre, Rayna Alexandre-Petrie (DSCI 4350, Group 5)
Shared at	GitHub: data/processed/; Google Drive: <ADD SHARED LINK>

Motivation and Intended Use

- Purpose: compare how well MRI scans, patient metadata, and their combination predict whether a patient's CDR will increase by the next visit.
- Intended users: the project team and course instructors, for academic model development and evaluation.
- Out-of-scope uses: clinical diagnosis, treatment decisions, or any individual-level medical use. The dataset is small, single-site, and not representative of the general population.

Sources and Lineage

Stage	Description
Original collection	OASIS-2: Open Access Series of Imaging Studies, longitudinal MRI data in nondemented and demented older adults (Marcus et al., 2010). Washington University Knight Alzheimer Disease Research Center.
Clinical source	Kaggle dataset "MRI and Alzheimers" (oasis_longitudinal.csv): 373 visits x 15 columns, 150 patients, 2 to 5 visits each. Used unmodified.
MRI source	OASIS-2 raw scan archives from oasis-brains.org: OAS2_RAW_PART1.tar.gz (10.9 GB, 209 sessions) and OAS2_RAW_PART2.tar.gz (8.4 GB, 164 sessions).
MRI build	src/data/make_mri_dataset.py: streams the archives and writes per-scan and per-session tables.
Dataset build	src/data/make_dataset.py: cleans the CSV, pairs visits, attaches the MRI session measures, splits, and imputes (deterministic, random_state = 42).
Versioning	Raw-file, archive, and output SHA-256 hashes are stored in oasis2_visit_pairs_metadata.json and oasis2_mri_metadata.json.

Composition

- Instance: one visit pair (visit t to visit t+1) for one patient. Patients with k visits contribute k - 1 pairs.
- Patients: 150 (41% male), ages 60 to 96 at the earlier visit (mean 76.1), all right-handed.
- Baseline CDR of pairs: 133 at CDR 0, 70 at CDR 0.5, 20 at CDR 1.
- Target balance: 192 stable, 31 declined. No patient declined more than once, so the 31 positives come from 31 different patients.

CDR transitions (rows = CDR at visit t, columns = CDR at visit t+1)

CDR at t	0	0.5	1	2
0	118	15	0	0
0.5	3	54	13	0
1	0	2	15	3

MRI scans

Property	Value
Sessions / scans	373 sessions (one per clinical visit), 1,367 scans; 2 to 4 repeated scans per session
Sequence	T1-weighted MPRAGE, sagittal acquisition, 1.5 T
Format and grid	Analyze 7.5 (.img + .hdr), big-endian float32; 256 x 256 x 128 voxels at 1 x 1 x 1.25 mm for every scan
Size	19.3 GB compressed, about 45 GB extracted (not stored in the repository)
Quality	SNR 6.7 to 36.4 (mean 21.4); 19 sessions flagged for motion (low repeat-scan agreement)

Fields (oasis2_visit_pairs.csv)

Column	Group	Description
subject_id	ID	Patient identifier (grouping key; not a feature)
mri_id	ID	MRI session of the earlier visit; joins to the MRI tables
visit	ID	Visit number of the earlier visit
age	Demographics	Age in years at the earlier visit
sex_male	Demographics	1 = male, 0 = female
educ	Demographics	Years of education
ses	Demographics	Socioeconomic status, 1 (highest) to 5 (lowest)
ses_imputed	Demographics	1 if SES was missing and imputed
mmse	Clinical	Mini-Mental State Examination, 0 to 30 (lower = worse)
cdr	Clinical	Clinical Dementia Rating at the earlier visit (0, 0.5, 1)
etiv	MRI (CSV)	Estimated total intracranial volume (cm^3)
nwbv	MRI (CSV)	Normalized whole-brain volume (fraction of eTIV)
asf	MRI (CSV)	Atlas scaling factor (about 1/eTIV; r = -0.99 with etiv)
days_since_baseline	Visit timing	Days since the patient's first visit
days_to_next	Visit timing	Days until the next visit (prediction horizon)
has_prev_visit	Visit timing	1 if the earlier visit has its own previous visit
days_since_prev	Visit timing	Days since the previous visit (0 if none)
mmse_change_prev	Change	MMSE change since the previous visit (0 if none)
nwbv_change_prev	Change	nWBV change since the previous visit (0 if none)
cdr_change_prev	Change	CDR change since the previous visit (0 if none)
mmse_rate_prev	Change	MMSE change per year since the previous visit (0 if none)
nwbv_rate_prev	Change	nWBV change per year since the previous visit (0 if none)

mri_csf_proxy	MRI scan	Share of dark (CSF-like) voxels in the central 40% box of the scan, averaged over repeats. Exploratory atrophy proxy; r = -0.54 with nWBV across the 373 sessions
mri_n_scans	MRI quality	Repeated scans in the session (2 to 4). Audit only
mri_snr	MRI quality	Signal-to-noise ratio (mean foreground / background SD). Audit only
mri_repeat_r	MRI quality	Pearson r between the first scan and each repeat; low = motion. Audit only
mri_qc_flag	MRI quality	1 if SNR or repeat-scan agreement is more than 3 IQR below Q1. Audit only
mmse_imputed	Audit	1 if MMSE was filled from the previous visit
cdr_next	Target source	CDR at the next visit. Defines the target; never use as a feature.
declined	Target	Target. 1 if cdr_next > cdr
split	Split	train or test (patient-level)
cv_fold	Split	CV fold 0 to 4 for training rows; -1 for test rows

oasis2_mri_sessions.csv (373 rows) and oasis2_mri_scans.csv (1,367 rows) hold the same MRI measures per session and per scan, plus header fields and raw intensity percentiles.

Preprocessing

Clinical data (make_dataset.py)

1. Validate the raw schema and check that MRI ID is unique.
2. MMSE (2 missing visits, 1 patient): filled with the same patient's previous MMSE. Only 1 of the resulting pairs is affected.
3. Change features: differences and yearly rates between each visit and the same patient's previous visit. They look backward only, so they contain no future information.
4. Pairing: each visit is paired with the next; the last visit of each patient is dropped (373 visits to 223 pairs). declined = 1 if CDR increases.
5. MRI join: the MRI session measures of visit t are attached by mri_id (all 223 pairs match).
6. Patient-level split: StratifiedGroupKFold holds out 1 of 5 patient groups as the test set (about 20%), stratified on the target; the training set gets 5 grouped, stratified CV folds.
7. SES (8 patients, 11 pairs): imputed with the mode of training patients (SES = 2); ses_imputed flags these rows.
8. Excluded raw columns: Group (assigned from the full visit history, so it leaks the outcome) and Hand (constant).
9. Encoding: M/F to sex_male; column names converted to snake_case.

MRI scans (make_mri_dataset.py)

1. Stream each archive once and decode every scan in memory with nibabel; nothing is extracted to disk. Archive SHA-256 hashes are computed during the same read.
2. Per scan: header fields, intensity percentiles above an Otsu threshold, SNR, and the CSF proxy.
3. Per session: average the scan metrics, measure repeat-scan agreement on 4x-downsampled volumes, and add the QC flags.
4. Link every session to the clinical visits and visit pairs and verify coverage.
5. Keep half-resolution central slices of the first scan per session (data/interim/, not versioned) for figures.

The scans are not yet skull-stripped, registered, or intensity-normalized; these steps belong to the CNN preprocessing in the modeling sprint.

Known Limitations and Biases

- Small sample: 223 pairs and 31 positives. The test set has only 6 positives, so test metrics will have wide uncertainty. Use the grouped CV folds for model selection.
- Class imbalance: 13.9% positives. Accuracy is misleading; report recall, F1-score, and ROC-AUC.
- Unequal follow-up: visit gaps range from 182 to 1,707 days. days_to_next is the horizon the prediction refers to and must be known (or planned) at prediction time.
- Coarse target: CDR moves in steps of 0.5. Five pairs show a CDR decrease and are labeled stable. Only 3 pairs reach CDR 2.
- Zero-filled change features: for the 150 first-visit pairs the change features are 0 by construction; has_prev_visit lets models tell these apart from true "no change".
- Redundant features: etiv and asf are almost perfectly correlated; linear models should keep only one.
- Raw MRI: scans include skull, neck, and face, are not registered, and have scanner-dependent intensity scales. The CSF proxy is a coarse measure, not a validated segmentation. Flagged sessions (mri_qc_flag = 1) are kept for a sensitivity analysis.
- Representativeness: one research site, right-handed volunteers aged 60+, no race or ethnicity information. Results may not generalize to other populations.

Ethical and Privacy Considerations

- OASIS data are de-identified. Do not attempt to re-identify participants or link them to other sources.
- Use must follow the OASIS Data Use Agreement and acknowledgment requirements published at https://www.oasis-brains.org/. Raw scans are not redistributed in the repository.
- Predictions from models trained on this data are research outputs, not medical advice.

Maintenance and Versioning

Rebuild in this order (outputs are deterministic for the same inputs):

1. python -m src.data.make_mri_dataset (about 2 to 5 minutes; needs both archives in data/raw/)
2. python -m src.data.make_dataset
3. python -m src.visualization.eda_report

Any change to the hashes, row counts, imputation values, or split statistics in the metadata JSON files requires a new dataset version.

Citation

Marcus, D. S., Fotenos, A. F., Csernansky, J. G., Morris, J. C., & Buckner, R. L. (2010). Open Access Series of Imaging Studies: Longitudinal MRI data in nondemented and demented older adults. Journal of Cognitive Neuroscience, 22(12), 2677-2684.